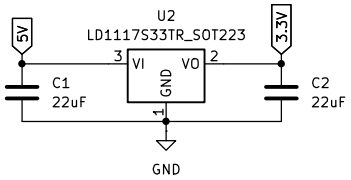
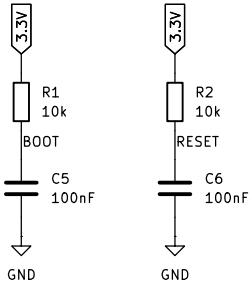
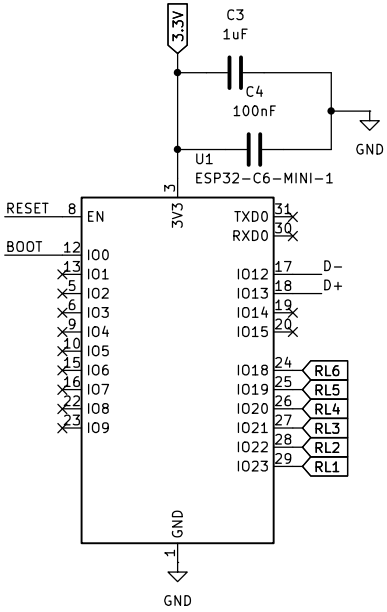
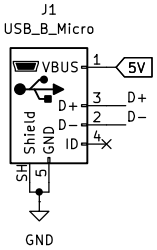
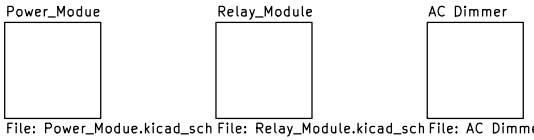
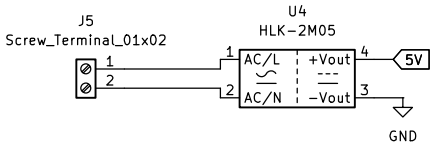

U2
LD1117S33TR_SOT223
5V
3.3V
VI
VO
GND
C1
22uF
C2
22uF
GND

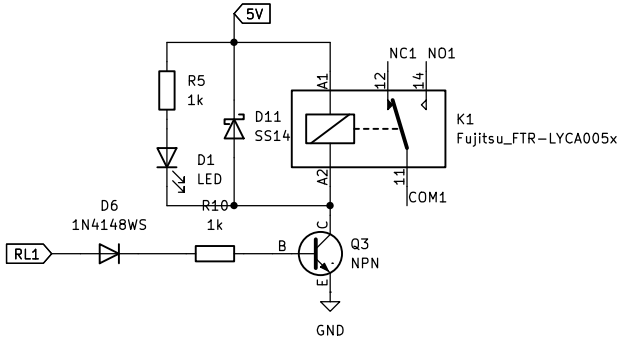

3.3V
R1
10k
BOOT
C5
100nF
GND
3.3V
R2
10k
RESET
C6
100nF
GND

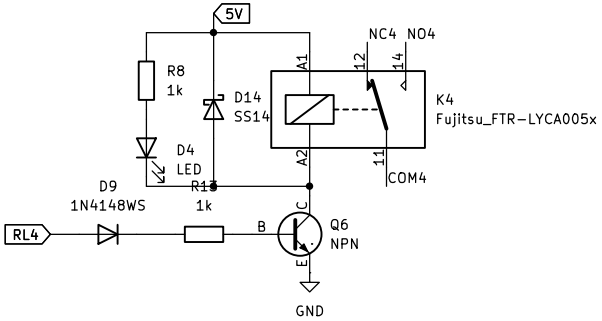

3.3V
C3
1uF
C4
100nF
GND
U1
ESP32-C6-MINI-1
RESET
BOOT
EN
3V3
GND
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
TXD0
RXD0
IO12
IO13
IO14
IO15
IO18
IO19
IO20
IO21
IO22
IO23
D-
D+
RL6
RL5
RL4
RL3
RL2
RL1
GND

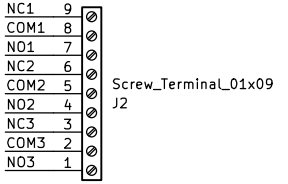

J1
USB_B_Micro
VBUS
5V
D+
D-
ID
GND
Shield
SH
D+
D-
GND

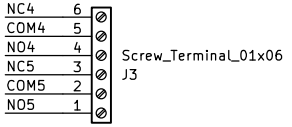

Power_Modue
Relay_Module
AC Dimmer
File: Power_Modue.kicad_sch
File: Relay_Module.kicad_sch
File: AC Dimmer.kicad_sch

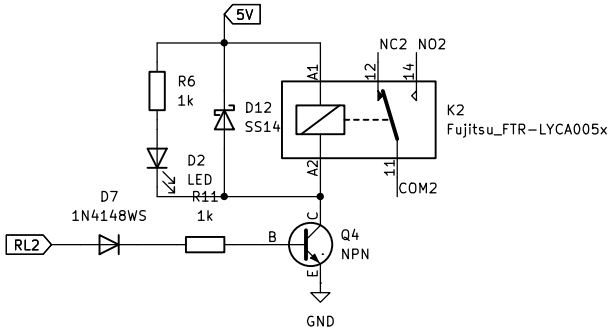

J5
Screw_Terminal_01x02
1
2
U4
HLK-2M05
1
AC/L
2
AC/N
+Vout
-Vout
4
3
5V
GND

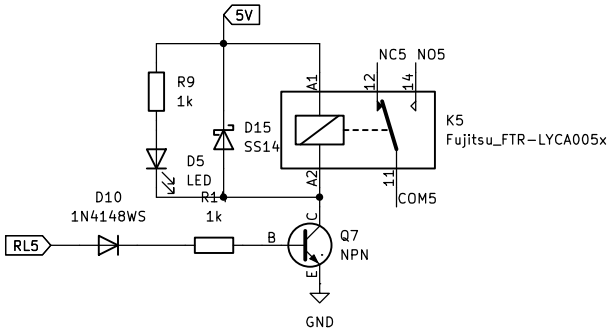

5V
R5
1k
D11
SS14
D1
LED
R10
1k
D6
1N4148WS
RL1
NC1
NO1
A1
12
14
A2
11
COM1
K1
Fujitsu_FTR-LYCA005x
B
C
E
Q3
NPN
GND

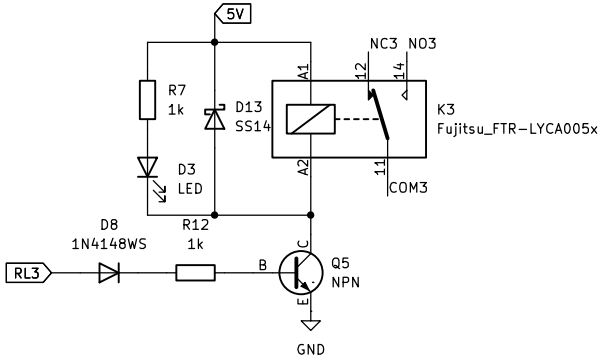

5V
R8
1k
D14
SS14
D4
LED
R13
1k
D9
1N4148WS
RL4
NC4
NO4
A1
12
14
A2
11
COM4
K4
Fujitsu_FTR-LYCA005x
B
C
E
Q6
NPN
GND

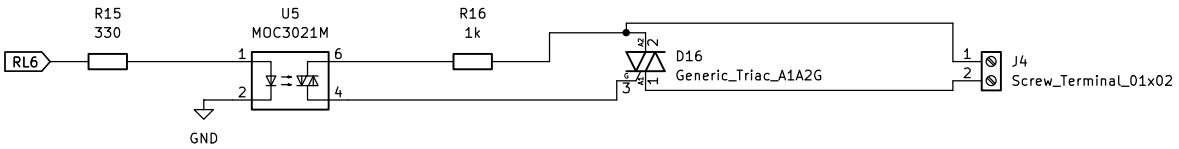

NC1 9
COM1 8
NO1 7
NC2 6
COM2 5
NO2 4
NC3 3
COM3 2
NO3 1
Screw_Terminal_01x09
J2


NC4 6
COM4 5
NO4 4
NC5 3
COM5 2
NO5 1
Screw_Terminal_01x06
J3


5V
R6
1k
D12
SS14
D2
LED
R11
1k
D7
1N4148WS
RL2
NC2
NO2
A1
12
14
A2
11
COM2
K2
Fujitsu_FTR-LYCA005x
B
C
E
Q4
NPN
GND


5V
R9
1k
D15
SS14
D5
LED
R14
1k
D10
1N4148WS
RL5
NC5
NO5
A1
12
14
A2
11
COM5
K5
Fujitsu_FTR-LYCA005x
B
C
E
Q7
NPN
GND


5V
R7
1k
D13
SS14
D3
LED
R12
1k
D8
1N4148WS
RL3
NC3
NO3
A1
12
14
A2
11
COM3
K3
Fujitsu_FTR-LYCA005x
B
C
E
Q5
NPN
GND


R15
330
U5
MOC3021M
R16
1k
RL6
1
2
6
4
GND
D16
Generic_Triac_A1A2G
3
1
2
J4
Screw_Terminal_01x02